Session 2 Labs — HTTP, your first page, Git

CM-501 Frontend Web Development · EPA 1

Team / names: _____

Lab A · Read real HTTP traffic (pairs, 30 min)

Setup: Chrome or Edge, **incognito window**, DevTools open (`F12` / `Cmd+Opt+I`), tab **Network**, tick **Disable cache**. Then load the page.

Site: `https://rathmayers.github.io/campus-cafe-demo/` (or your local copy `http://localhost:4321/`)

A1 · The main document (≈ 8 min)

Open `…/after/`. In the Network list click the **first** request (the HTML document).

Question	Your answer
Request URL and method	https://rathmayers.github.io/campus-cafe-demo/after/ and GET
Status code	200
Protocol (right-click the column header → enable **Protocol**: h2? h3?)	h2
Remote address (IP:port)	185.199.109.153:443
`content-type` of the response	text/html; charset=utf-8
`cache-control` of the response	max-age=600
`etag` or `last-modified` present?	Yes
One request header that tells the server something about you	User Agent: tells about my browser

A2 · Before vs after (≈ 10 min)

Load `…/before/` and `…/after/` one after the other. Read the bottom bar of the Network tab.

	before	after
Number of requests	12	8

	before	after
Transferred (MB / kB)	22.6 MB	53.0 kB
`DOMContentLoaded`	151 ms	26 ms
`Load`	620 ms	43 ms
Largest file (name + size)	Hero PNG, 7.0 MB	Atkinson font, 17.8 kB
Number of different domains (add column **Domain**)	4	1
Same with throttling **Slow 4G**: `Load`	1.3 min	1.22 s

Which three requests on *before* would you remove first? Why?

I would remove the 7 MB hero image
Remove unnecesarry scripts
Remove unused fonts

A3 · Provoke status codes (≈ 6 min)

Try this	Status you see	What it means
`…/campus-cafe-demo/does-not-exist/`	404	Not Found
`http://rathmayers.github.io/campus-cafe-demo/` (with **http**) – look at the first request	301	Moved Permanently
Reload `…/after/` **without** "Disable cache"	304	not Modified

A4 · The terminal view (≈ 6 min)

macOS / Linux: `curl -I https://rathmayers.github.io/campus-cafe-demo/after/` Windows PowerShell: `curl.exe -I https://rathmayers.github.io/campus-cafe-demo/after/`

Copy three header lines and explain them in your own words:

1. ETag: "6acd7532-1a02" Identifies the version of the file.
2. Cache-Control: max-age=600 Tells the browser it can cache the file for 10 minutes.
3. Last-Modified: Tue, 13 Oct 2026 00:02:58 GMT Shows when the file was last modified.

If you're fast: `curl -I http://rathmayers.github.io/campus-cafe-demo/` (http!) — which status and which `location`? Compare the response headers of `…/after/` with those of a big site (e.g. `https://www.gov.uk`): which security headers does each set?

Lab B · Your first page, by hand (individual, 20 min)

Rules: no framework, no template, no copy-paste from AI. Type it yourself: you must be able to explain every line.

1. Create a folder `cm501-<yourname>` (e.g. in `Documents` or `~/Development`) and open it in **VS Code** (*File → Open Folder*).
2. Create `index.html` with this skeleton and fill it in:

```
<!doctype html>
<html lang="en">
  <head>
    <meta charset="utf-8">
    <meta name="viewport" content="width=device-width, initial-scale=1">
    <title>Your Name · CM-501</title>
  </head>
  <body>
    <h1>Hi, I'm …</h1>
    <p>One or two sentences about you and what you want to learn.</p>
    <h2>Websites I love and hate</h2>
    <ul>
      <li><a href="https://…">Name</a>: why (with evidence)</li>
    </ul>
  </body>
</html>
```

3. Serve the folder: open the VS Code terminal (`Ctrl+`) and run `npx --yes serve .` → open the URL shown (usually `http://localhost:3000`). *No Node.js? Open the file directly by double-click; that works for today.*
4. DevTools **Elements**: find your `<h1>` in the DOM tree, double-click its text and change it. Reload: the change is gone. *Why?* _____
5. DevTools **Network**: how many requests does your page need? _____ Why so few? _____
6. Create `about.html` (same skeleton, different content) and link the pages to each other: `<a href="about.html">About me</a>`.

If you're fast: add `<script src="slow.js"></script>` in the `<head>` with `slow.js` containing `const t = Date.now(); while (Date.now() - t < 3000) {}` and reload. What happens? Then add `defer`. What changes? (Remove it again afterwards!)

Lab C · Ship it to GitHub (individual, 18 min)

Once per computer (skip if done before; use the e-mail of your GitHub account):

```
git config --global user.name "Your Name"
git config --global user.email "you@example.com"
git config --global init.defaultBranch main
```

Steps (terminal in your `cm501-<yourname>` folder):

#	Do	Command
1	Start a repository	`git init`
2	See what Git sees	`git status`
3	Commit 1: the skeleton	`git add index.html` → `git commit -m "Add page skeleton"`
4	Commit 2: your content	edit → `git add index.html` → `git commit -m "Add intro and website list"`
5	Commit 3: about page	`git add about.html index.html` → `git commit -m "Add about page and navigation links"`
6	Look at your history	`git log --oneline`
7	On github.com: **New repository** `cm501-<yourname>`, **public**, **no** README, **no** .gitignore	
8	Connect and upload	`git remote add origin https://github.com/<you>/cm501-<yourname>.git` → `git push -u origin main`
9	Check	refresh the repository page on github.com: 3 commits?

Typical problems

Message	Fix
`Author identity unknown`	run the `git config` lines above
`src refspec main does not match any`	you have no commit yet, or your branch is `master`: `git branch -M main`
`rejected … fetch first`	the GitHub repo is not empty (README): `git pull --rebase origin main`, then push again
Login fails / password not accepted	use the browser login window, or create a *personal access token* on GitHub (Settings → Developer settings) and use it as password

If you're fast: add a `README.md` (what is this repo? how to view it?) and a `.gitignore` containing `.DS_Store` and `node_modules/`. Commit both with good messages.